Anant Khanna

Design Verification | SystemVerilog | FPGA | GPU Compute
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Education

University of British Columbia Vancouver, BC
Bachelor of Applied Science in Computer Engineering (Co-op) - Dean's List 2025 *Expected Graduation: April 2029*

- **Relevant Coursework**: Computing Systems I & II (**SystemVerilog** digital design, **computer architecture**, caches, C), Operating Systems, Algorithms & Data Structures, Circuit Analysis

Experience

AMD | *Incoming Design Verification Engineering Intern* May 2027 – Dec. 2027

- Selected for an **8-month** co-op on the CIT (Chiplet Interconnect Technologies) **design verification** team: **SystemVerilog/UVM** test development, register testing, and **regression** infrastructure and automation

UBC UAS | *Machine Learning Team Lead* Sept. 2024 – Present

- Integrated YOLOv11 (**PyTorch**) pipeline in **Python** and **C++**, enabling real-time autonomous object detection on an **NVIDIA Jetson Orin Nano** travelling at **50+** km/h
- Implemented **TensorRT** FP16 quantization to compress model weights, reducing memory bandwidth usage by **50%** while maintaining inference precision within **0.5%** of the baseline

Agriprix | *Founding Software Engineer* May 2026 – Aug. 2026

- Led a platform security audit surfacing **67** vulnerabilities (5 critical), patching mass-assignment and IDOR flaws in Django REST Framework serializers
- Eliminated overselling **race conditions** via row-level locking and shipped a 15-step end-to-end Playwright **test** journey covering the supplier-to-buyer order flow

Projects

GPU Inference Benchmark | *Python, PyTorch, CUDA/TensorRT, ROCm/MIGraphX* July 2026 – Aug. 2026

- Built a three-rung benchmark ladder (**PyTorch** to **TensorRT** FP32 to TensorRT FP16) that holds precision constant across stages to isolate graph-compilation gains from quantization gains, benchmarking YOLO11n on an **RTX 3080** across **100** timed runs per config
- Applied rigorous methodology: 50 discarded warmup runs, `torch.cuda.synchronize()` to avoid timing async GPU dispatch, and median/p95 latency reported against a measured ±0.5ms noise floor
- Ported the harness to AMD MI300X via **ROCm/MIGraphX** (PyTorch FP32: **1,143** fps vs. **1,008** fps TensorRT FP16 on the 3080), then scaled the FP16 engine from batch 1 to 8 with near-zero latency penalty (1.03ms to 1.10ms) to **7,277** fps

Digital Systems Design (CPEN 211) | *SystemVerilog, QuestaSim, Quartus* Sept. 2025 – Dec. 2025

- Designed a CPU datapath and control (ALU, register file, control **FSM**) and a keyboard interface with a two-flop **synchronizer** and debounce logic in **SystemVerilog**, synthesized to a DE10-Lite FPGA with Intel Quartus
- Wrote self-checking, vector-driven **testbenches** in **QuestaSim** with golden vectors loaded from file, stimulus driven and checked on opposite clock edges to avoid races, four-state comparison, and pass/fail error reporting

FPGA-Based Neural Network Inference Accelerator | *SystemVerilog, C++, ARM* July 2025 – Sept. 2025

- Designed a single-layer neural network (MAC unit) on a DE1-SoC FPGA in **SystemVerilog**, pipelined into three stages (load, multiply, accumulate) to shorten the critical path, with 8-bit inputs, 16-bit products and a 24-bit accumulator sized from bit-growth analysis
- **Verified** against a reference model with a self-checking **testbench** covering edge cases; found and fixed a pipeline-drain bug by **waveform analysis** of valid-in/valid-out timing and adding a drain state
- Developed a memory-mapped **C++** driver on the ARM HPS under embedded Linux, achieving **3,000** inferences/sec and a **300%** speedup over ARM-only execution by optimizing HPS–FPGA communication

Technical Skills

Hardware/RTL: SystemVerilog, Verilog, FPGA Design (Intel Quartus, QuestaSim, Platform Designer), Digital Design, Computer Architecture, ARM SoCs
Verification: Self-Checking Testbenches, Reference Models, Simulation & Waveform Debugging, Corner-Case Tests, UVM Fundamentals
Languages & Tools: C, C++, Python, SystemVerilog, SQL, RISC-V & ARM Assembly, Linux, Git, Docker
GPU/ML: CUDA/TensorRT, ROCm/MIGraphX, PyTorch, TensorFlow, YOLO/Ultralytics